

Statement of Work/Requirements Document
Title: Dilution Refrigerator Magnet for Quantum Memory

I. Introduction/Background

In collaboration with the University of Chicago and Carnegie Mellon University, the National Institute of Standards and Technology (NIST) Information Technology Laboratory (ITL) is developing a silicon carbide (SiC) microcavity-enhanced quantum memory based on erbium-doped yttrium orthosilicate (Er:YSO). Quantum memory is a critical component for quantum communication and networking systems, particularly in enabling quantum repeaters and distributed quantum network nodes.

The SiC microcavity-enhanced Er:YSO platform represents a promising approach for achieving high-efficiency quantum memory at telecommunications wavelengths. As part of the Quantum Communication and Networking (QuCaN) project within NIST ITL, this effort aims to advance the performance and integration of quantum memory technologies in practical network environments.

Purpose/Objective: The purpose of this requirement is to procure a three-dimensional (3D) vector magnet system compatible with an existing dilution refrigerator. Operation of the quantum memory requires millikelvin (mK) cryogenic temperatures in conjunction with a precisely controlled 3D vector magnetic field. ITL currently operates a dilution refrigerator capable of reaching approximately 10 mK; however, it is not equipped with an integrated magnet system. The resulting system will support experimental validation and integration of quantum memory within NIST quantum networking testbeds, including the NIST Gaithersburg (NG-QNet) and the Washington, DC metropolitan area (DC-QNET) testbeds located on the NIST Gaithersburg campus.

II. Scope

The Contractor shall deliver a quantity of one (1) one 3D vector magnet system inclusive of FOB Destination delivery, installation, warranty, and training (as detailed herein).

III. Minimum Requirements

The Contractor shall provide a system that meets all technical specifications identified below. All items must be new. Used or remanufactured equipment will not be considered for award. Experimental, prototype, or custom items will not be considered. The use of “gray market” components are not authorized for sale in the U.S. by the Contractor is not acceptable. All line items shall be shipped in the original manufacturer’s packaging and include all original documentation and software, when applicable.

Line Item 0001:

Description: 3D Vector Magnet with Power Supply and Programmer

Quantity: One (1) Lot

A. Technical Specifications

- a. Rated Field(Z-Y-X): 6-1-1 Tesla at 4.2K
- b. Homogeneity(Z-Y-X): +/-0.5%(Z), +/-1%(Y), +/-1.5%(X)
- c. Persistent Current Switch to enable charging superconducting magnets without continuous power supply.
- d. Three axis Power Supply and Programmer to drive and manage the 3D vector superconducting magnet system
- e. Must be fully compatible with Bluefors LD 250 Dilution Refrigerator

IV. Deliverables

<i>Description</i>	<i>Quantity or Format</i>	<i>Due Date</i>
3D Vector Magnet with Power Supply and Programmer	1 Lot	Lead Time/NLT 8 months after receipt of offer

V. Delivery Terms

Delivery shall be F.O.B Destination and shall occur in accordance with the delivery due date provided in the above table.

FOB Destination means: The contractor shall pack and mark the shipment in conformance with carrier requirements, deliver the shipment in good order and condition to the point of delivery specified in the purchase order, be responsible for any loss of and/or damage to the goods occurring before receipt and acceptance of the shipment by the consignee at the delivery point specified in the purchase order; and pay all charges to the specified point of delivery. The contractor shall deliver all Line Items to:

National Institute of Standards and Technology
Shipping and Receiving
100 Bureau Drive, Building 301
Gaithersburg, MD 20899
COR: Eboni Harbeson; Building: 225

*****Due to ongoing limited site access of the NIST facilities, the Awardee SHALL confirm with the Designated Government Official and the Contracting Officer prior to ANY shipment to NIST*****

VI. Inspection and Acceptance:

In addition to the inspection and acceptance terms articulated in 52.212-4, the Government reserves the right to perform such performance tests and evaluations as defined below to verify specified system performance. Such tests and evaluations, if performed, shall be conducted within the environment that the system is to be operated. Upon delivery of device, the Government will perform the test within one month. The Contractor has the right to be present during the tests and evaluations, if performed, at the Contractor's expense.

The scope of general inspection included:

- Structural, mechanical, and electrical evaluation of product
- An operational evaluation of the time tagging unit
- Evaluation of the system's performance and software interfacing

A visual inspection of the laser will be performed by NIST to identify surface defects or any form of indication that the time tagging unit was damaged during transport to NIST. The Government shall have sole discretion to require repair or replacement of damaged and/or nonconforming supplies at no cost to the Government. The Government at any time prior to acceptance shall reject the time tagging unit due to defects and/or nonconformance.

The contractor is responsible for latent defects discovered any time after final inspection. However, the extent of its liability shall be prorated over the useful life of the time tagging unit.

The place of acceptance shall be:

100 Bureau Drive
Building 222
Gaithersburg, MD 20899

Ownership of the tunable laser unit shall transfer to NIST upon acceptance.

VII. Warranty

The contractor shall warrant the entire system in accordance with the Original Equipment Manufacturers' (OEM) standard commercial item warranty and shall be in accordance with terms in FAR 52.212-4. Warranty shall commence upon acceptance of the system by the Government and at a minimum shall include parts and labor.

VIII. Payment Schedule

The Contractor shall be paid, in accordance with Net 30-day payment terms, upon receipt and acceptance of a proper invoice, in accordance with the following schedule:

1. 100% after receipt and acceptance by NIST of fully delivered system.
2. The Government anticipates inspection will occur upon after the testing procedures set forth in this document have been complete

NOTE: Partial shipments and partial invoices will not be accepted, unless otherwise requested and accepted by the Contracting Officer prior to award offer.